

$$115) (5-11y)-(-3y^2+9-13y)$$

$$5-11y+3y^2-9+13y$$

$$3y^2+(-11y+13y)+(-9+5)$$

$$\underline{3y^2+2y-4}$$

$$116) (-14x^4+3x^5)+(-12x^4+4x+2x^5)$$

$$-14x^4+3x^5-12x^4+4x+2x^5$$

$$-14x^4-12x^4+4x+(3x^5+2x^5)$$

$$\underline{-26x^4+4x+5x^5}$$

$$117) (4b^2+7b^4+12b-7)+(11+7b+4b^2)+(10b^4+b)$$

$$4b^2+7b^4+12b-7+11+7b+4b^2+10b^4+b$$

$$(7b^4+10b^4)+(4b^2+4b^2)+(12b+7b+b)+(-7+11)$$

$$\underline{17b^4+8b^2+20b+4}$$

$$118) (3-2x-9x^4-8x^5)-(1+9x^5+6x)+(13x^2+8x^5)$$

$$3-2x-9x^4-8x^5-1-9x^5-6x+13x^2+8x^5$$

$$-8x^5-9x^4+8x^5-9x^4+13x^2-2x-6x+3-1$$

$$(-8+8)x^5-9x^4+13x^2-8x+2$$

$$\underline{-9x^4+13x^2-8x+2}$$

$$119) (11b^4+5b^2+8+9b^3)+(6b^4-8b^2-7)+(-13b^3+11b^4)$$

$$11b^4+5b^2+8+9b^3+6b^4-8b^2-7-13b^3+11b^4$$

$$(11b^4+6b^4+11b^4)+(9b^3-13b^3)+(5b^2-8b^2)+(8-7)$$

$$\underline{28b^4-4b^3-3b^2+1}$$

$$120) (-11h^2+9h^5+11h^4+12h)-(9h^5+7+8h^4)+(-2h^2-14h^5)$$

$$-11h^2+9h^5+11h^4+12h-9h^5-7-8h^4-2h^2-14h^5$$

$$(9h^5-9h^5-14h^5)+(11h^4-8h^4)$$

$$+(-11h^2-2h^2)+12h-7$$

$$\underline{-14h^5-13h^2+12h-7}$$

$$121) -3b(-b^2-5b-2) \quad (-3b)(-b^2)+(-3b)(-5b)+(-3b)(-2)$$

$$\underline{3b^3+15b^2+6b}$$

$$122) 3h(-3h^2-h+4) \quad (3h)(-3h^2)+(3h)(-h)+(3h)(4)$$

$$\underline{-9h^3-3h^2+12h}$$

$$123) 8(-7m^2-5m-8) \quad 8(-7m^2)+8(-5m)+8(-8)$$

$$\underline{-56m^2-40m-64}$$

$$124) -8(6x^2+4x+5) \quad (-8)(6x^2)+(-8)(4x)+(-8)(5)$$

$$\underline{-48x^2-32x-40}$$

$$125) 2(2x^2-6x-4) \quad (2)(2x^2)+(2)(-6x)+(2)(-4)$$

$$\underline{4x^2-12x-8}$$

$$126) -7r^2(-3r^2+3r-8) \quad (-7r^2)(-3r^2)+(-7r^2)(3r)+(-7r^2)(-8)$$

$$\underline{21r^4+21r^3+56r^2}$$

$$127) (8h-6)(3h^2-6h+8) \quad \begin{array}{ll} (8h)(3h^2)=24h^3 & (-6)(3h^2)=-18h^2 \\ (8h)(-6h)=-48h^2 & (-6)(-6h)=36h \\ (8h)(8)=64h & (-6)(8h)=-48 \end{array}$$

$$24h^3-48h^2+64h-18h^2+36h-48 \quad \underline{24h^3-66h^2+100h-48}$$

$$128) (6n+3)(5n^2-3n-5) \quad \begin{array}{ll} 6n(5n^2)=30n^3 & 3(5n^2)=15n^2 \\ 6n(-3n)=-18n^2 & 3(-3n)=-9n \\ 6n(-5)=-30n & 3(-5)=-15 \end{array}$$

$$30n^3-18n^2-30n+15n^2-9n-15 \quad \underline{30n^3-3n^2-39n-15}$$

$$129) (8b+3)(5b^2+2b-5) \quad \begin{array}{ll} 8b(5b^2)=40b^3 & 3(5b^2)=15b^2 \\ 8b(2b)=16b^2 & 3(2b)=6b \\ 8b(-5)=-40b & 3(-5)=-15 \end{array}$$

$$40b^3+16b^2-40b+15b^2+6b-15 \quad \underline{40b^3+31b^2-34b-15}$$

$$130) (8x-7)(2x^2-x-6) \quad 8x(2x^2)=16x^3 \quad -7(2x^2)=-14x^2 \\ 8x(-x)=-8x^2 \quad -7(-x)=7x \\ 8x(-6)=-48x \quad -7(-6)=42$$

$$16x^3-8x^2-48x-14x^2+7x+42 \quad \underline{16x^3-22x^2-41x+42}$$

$$131) (5x-7)(2x^2+3x+3) \quad 5x(2x^2)=10x^3 \quad -7(2x^2)=-14x^2 \\ 5x(3x)=15x^2 \quad -7(3x)=-21x \\ 5x(3)=15x \quad -7(3)=-21$$

$$10x^3+15x^2+15x-14x^2-21x-21 \quad \underline{10x^3+x^2-6x-21}$$

$$132) (5x-2)(8x^2+7x-2) \quad 5x(8x^2)=40x^3 \quad -2(8x^2)=-16x^2 \\ 5x(7x)=35x^2 \quad -2(7x)=-14x \\ 5x(-2)=-10x \quad -2(-2)=4$$

$$40x^3+35x^2-10x-16x^2-14x+4 \quad \underline{40x^3+19x^2-24x+4}$$

$$133) \frac{a^2 \cdot (3a)^2}{2a^0}, (3a)^2 = 3^2 a^2 = 9a^2, a^0 = 1, \frac{a^1(9a^2)}{2(1)}, \frac{9a^{2+2}}{2} \quad \underline{\frac{9a^4}{2}}$$

$$134) \left(\frac{3r^3 \cdot 2r^3}{2r^2} \right)^2 \quad 3r^3 \cdot 2r^3 = 6r^6 \quad \left(\frac{6r^6}{2r^2} \right)^2 \quad (3r^{6-2})^2 \quad (3r^4)^2 \quad \underline{9r^8}$$

$$135) \frac{v^0 \cdot (3v)^2}{(v^0)^3} \quad v^0 = 1 \quad (3v)^2 = 9v^2 \quad \frac{1 \cdot 9v^2}{1^3} \quad \underline{9v^2}$$

$$136) \left(\frac{2r^3}{2r^2 \cdot 3v} \right)^3 \quad 2r^2 \cdot 3v = 6r^2v \quad \left(\frac{2r^3}{6r^2v} \right)^3 \quad \left(\frac{1}{3} \right)^3 \quad \underline{\frac{1}{27}}$$

$$137) \left(\frac{(3x)^3}{3x \cdot 3x} \right)^2 \quad (3x)^3 = 27x^3 \quad 3x \cdot 3x = 9x^2 \quad \left(\frac{27x^3}{9x^2} \right)^2 \quad (3x)^2 \quad \underline{9x^2}$$

$$138) \frac{x^2 \cdot (2x^3)^2}{3x} \quad (2x^3)^2 = 4x^6 \quad \frac{x^2 \cdot 4x^6}{3x} \quad \frac{4x^8}{3x} \quad \frac{4}{3} x^{8-1} \quad \underline{\frac{4x^7}{3}}$$

$$139) \sqrt{18x^3} \quad 18x^3 = 9x^2 \cdot 2x \quad \sqrt{9x^2 \cdot 2x} \quad \underline{3x\sqrt{2x}}$$

$$140) \sqrt[4]{80h^6} \quad 80h^6 = 16h^4 \cdot 5h^2 \quad \sqrt[4]{16h^4 \cdot 5h^2} \quad 2h \sqrt[4]{5h^2}$$

$$141) \sqrt{192h^3} \quad 192h^3 = 64h^2 \cdot 3 \quad \sqrt{64h^2 \cdot 3h} \quad 8h\sqrt{3h}$$

$$142) \sqrt{288h} \quad 288 = 144 \cdot 2 \quad \sqrt{144 \cdot 2h} \quad 12\sqrt{2h}$$

$$143) \sqrt{18v^2} \quad 18v^2 = 9v^2 \cdot 2 \quad \sqrt{9v^2 \cdot 2} \quad 3v\sqrt{2}$$

$$144) \sqrt[4]{32h^4} \quad 32h^4 = 16h^4 \cdot 2 \quad \sqrt[4]{16h^4 \cdot 2} \quad \sqrt[4]{16h^4} \cdot \sqrt[4]{2} \quad \sqrt[4]{32h^4}$$

$$145) \frac{2}{5+3\sqrt{3}} \cdot \frac{5-3\sqrt{3}}{5-3\sqrt{3}} = 2(5-3\sqrt{3}) = 10-6\sqrt{3} = (5+3\sqrt{3})(5-3\sqrt{3})$$

$$(a+b)(a-b) = a^2 - b^2 \quad 5^2 - (3\sqrt{3})^2 \quad 25 - 9(3) \quad 25 - 27 = -2$$

$$\frac{10-6\sqrt{3}}{-2} \quad \frac{5+3\sqrt{3}}{1}$$

$$146) \frac{4}{5\sqrt{3}-4} \cdot \frac{5\sqrt{3}+4}{5\sqrt{3}+4} \quad \frac{4}{5\sqrt{3}-4} \cdot \frac{5\sqrt{3}+4}{5\sqrt{3}+4} \quad 4(5\sqrt{3}+4) = 20\sqrt{3}+16$$

$$(5\sqrt{3})^2 - 4^2 \quad 25(3) - 16 \quad 75 - 16 = 59 \quad 20\sqrt{3}+16$$

$$\frac{59}{1}$$

$$147) \frac{2}{3\sqrt{2}+5\sqrt{3}} \cdot \frac{3\sqrt{2}-5\sqrt{3}}{3\sqrt{2}-5\sqrt{3}} \quad \frac{2}{3\sqrt{2}+5\sqrt{3}} \cdot \frac{3\sqrt{2}-5\sqrt{3}}{3\sqrt{2}-5\sqrt{3}}$$

$$2(3\sqrt{2}-5\sqrt{3}) = 6\sqrt{2}-10\sqrt{3} \quad (3\sqrt{2})^2 - (5\sqrt{3})^2 \quad 9(2) - 25(3)$$

$$18 - 75 = -57$$

$$\frac{6\sqrt{2}-10\sqrt{3}}{-57}$$

$$148) \frac{4}{\sqrt{3}+2} \cdot \frac{\sqrt{3}-2}{\sqrt{3}-2} \cdot \frac{\sqrt{3}+2}{\sqrt{3}+2} \quad 4(\sqrt{3}-2) = 4\sqrt{3}-8$$

$$4(\sqrt{3}-2) = 4\sqrt{3}-8 \quad (\sqrt{3})^2 - 2^2 \quad 3-4 = -1 \quad \frac{4\sqrt{3}-8}{-1}$$

$$-4\sqrt{3}+8$$

$$149) \frac{3}{-4+4\sqrt{2}} \cdot 4(\sqrt{2}-1) \cdot \frac{3}{4(\sqrt{2}-1)} \cdot \frac{\sqrt{2}+1}{\sqrt{2}+1} \quad \frac{3(\sqrt{2}+1)}{4(\sqrt{2}-1)(\sqrt{2}+1)}$$

$$(\sqrt{2})^2 - 1^2 \quad 2-1 = 1 \quad \frac{3(\sqrt{2}+1)}{4}$$

$$150) \frac{2}{4+\sqrt{2}} \cdot 4-\sqrt{2} \cdot \frac{2}{4+\sqrt{2}} \cdot \frac{4-\sqrt{2}}{4-\sqrt{2}} \quad 2(4-\sqrt{2}) = 8-2\sqrt{2}$$

$$4^2 - (\sqrt{2})^2 \quad 16-2 = 14 \quad \frac{8-2\sqrt{2}}{14} \quad \frac{4-\sqrt{2}}{7}$$